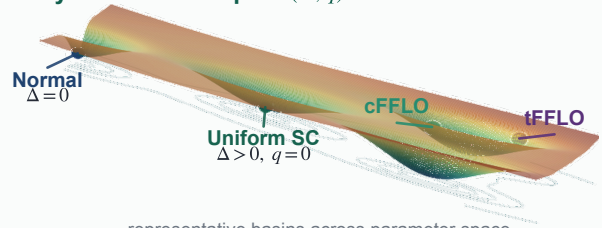


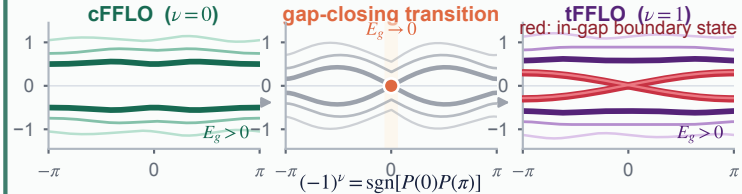
Physical state space generated by the BdG oracle

Thermodynamic landscape $F(\Delta, q)$

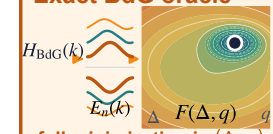


representative basins across parameter space

Topology resolution ($\mathbb{Z}_2$ invariant)



Exact BdG oracle



full minimization in (Δ, q)

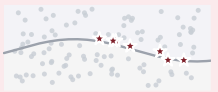
Reliability gate

- ✓ optimization convergence
- ✓ spectral/topological trust
- ✓ numerical reliability

Physical observation

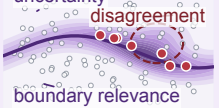
Phase	FFLO
Order	Δ, q
Energy	F^*
Spectrum	E_g
Topology	ν
Uncertainty	σ_{num}

Next exact calls



selected micro-batch

Acquisition



$A(x) = f[\sigma(x), d_{\mathrm{boundary}}, \Delta F]$

Surrogate scheduler

